
Advanced Computer Architectures

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Name
Last Name
ID Number (codice persona)

Problem 1 (20%)	
Problem 2 (15%)	
Problem 3 (15%)	
Question 1 (15%)	
Question 2 (20%)	
Question 3 (15%)	
Total (100%)	

NOTE: students have 90' to complete the exam.

To pass the midterm, a minimum score of 20% is requested by each “section” (section 1: problems, section 2: questions).

The overall score to access the oral has to be greater or equal to 50%

Problem 1 (20 %)

Consider the following assembly executed to be executed on a CPU with dynamic scheduling based on TOMASULO with a ROB. Consider an architecture with:

- 4 RESERVATION STATIONS (RS1, RS2, RS3, RS4) + 2 LDU units (LDU1, LDU2) with latency 2 cycles
- 2 RESERVATION STATION (RS5, RS6) + 2 FPU (FPU1, FPU2) with latency 3 cycles
- 3 RESERVATION STATIONS (RS7, RS8) + 1 Integer ALU (ALU1) with latency 1 cycle
- 10-slots ROB (from 0 to 9)
- Separated CDB for integer and floating-point operations
- Assume the branch is wrongly speculated as taken. Consider the misprediction resolution to take 10 cycles on top of the ALU latency.

Assembly:

```
FOR: LD $F2, A($R1)
I1: LD $F4, B($R1)
I2: LD $F6, C($R1)
I3: FADD $F4, $F2, $F4
I4: FADD $F6, $F4, $F6
I5: SD $F6, A($R1)
I6: ADDUI $R1, $R1, 8
I7: BNE $R1, $R2, FOR
```

P1.A List all the possible true data dependencies and control dependencies in the code considering two iterations executing.

Answer P1.A

P1.B Execute the two iterations of the previous code, according to the wrong speculation prediction assumption and dependencies highlighted before.

P1.C Report the structural details of the architecture (i.e., rob slots, rs and fu numbers) in the table.

Answer P1.B && P1.C

Instruction	ISSUE	START EXE	EXE COMPLETE	WB	Commit	ROB idx	RSi	Unit
IT1 - FOR: I0: LD \$F2, A(\$R1)								
IT1 - I1: LD \$F4, B(\$R1)								
IT1 - I2: LD \$F6, C(\$R1)								
IT1 - I3: FADD \$F4, \$F2, \$F4								
IT1 - I4: FADD \$F6, \$F4, \$F6								
IT1 - I5: SD \$F6, A(\$R1)								
IT1 - I6: ADDUI \$R1, \$R1, 8								
IT1 - I7: BNE \$R1, \$R2, FOR								
IT2 - FOR: I0: LD \$F2, A(\$R1)								
IT2 - I1: LD \$F4, B(\$R1)								
IT2 - I2: LD \$F6, C(\$R1)								
IT2 - I3: FADD \$F4, \$F2, \$F4								
IT2 - I4: FADD \$F6, \$F4, \$F6								
IT2 - I5: SD \$F6, A(\$R1)								

Problem 2 (15%)

Assume that the following code has been executed on a CPU with Scoreboard with an extended physical register file of 128 physical registers. The PRF has no limitations and can accommodate a parallel infinite amount of reads and writes.

	Instruction	ISSUE	READ OPERAND	EXE COMPLETE	WB
1	ld \$f1, FORCE(\$fp)	1	2	4	5
2	ld \$f0, DY(\$fp)	2	3	5	6
3	mul.d \$f0, \$f1, \$f0	3	7	10	11
4	ld \$f1, FY(\$fp)	4	5	7	8
5	add.d \$f0, \$f1, \$f0	5	12	15	16
6	sd \$f0, FY(\$fp)	6	17	19	20
7	ld \$f1, FORCE(\$fp)	7	8	10	11
8	ld \$f0, DZ(\$fp)	8	9	11	12
9	mul.d \$f0, \$f1, \$f0	9	13	16	17
10	ld \$f1, FZ(\$fp)	10	11	13	14
11	add.d \$f0, \$f1, \$f0	11	18	21	22
12	sd \$f0, FZ(\$fp)	12	23	25	26
13	lw \$2, 28(\$fp)	15	16	18	19
14	addi \$2, \$2, 1	16	20	21	22
15	sw \$2, 28(\$fp)	17	23	25	26

P2.A List all the possible true data dependencies – and only them - in the code.

Answer P2.A

P2.B Is there a Scoreboard “configuration” (i.e., FUs numbers and write ports) that can respect the shown execution? Yes or No? Please explain why, and, if yes, please detail the architecture characteristics.

Assume that all the memory operations are certainly without any ambiguity.

Answer P2.B

P2.C Consider the previous execution (regardless it was a possible execution or not), report the rename table status at cycle 8 and 15 considering physical registers of 64 bits. Explain your choice and assumptions

Answer 2.C

Renaming Table	Cycle 0	
F0	F1	\$2
P0	P2	P130

Renaming Table	Cycle 8	
F0	F1	\$2

Renaming Table	Cycle 15	
F0	F1	\$2

Problem 3 (15%)

Describe (the answer has to be effectively supported) a 1-BHT and a 2-BHT able to execute the following assembly code

```
    li t0, 0
outer_loop:
    li    t6, 2
    mul   t3, t0, t6
    li t1, 0
inner_loop:
    add t4, t0, t1
    addi t1, t1, 1
    li    t2, 100
    blt   t1, t2, inner_loop
    mul   t5, t0, t0
    addi t0, t0, 1
    li    t2, 1000
    blt   t0, t2, outer_loop
```

The obtained result, in terms of miss predictions, is in line with theoretical characteristics of the two predictors? Please effectively support your answer.

P3 Answer

P3 Answer (continued)